Symmetry-Quotient Forgery Attacks on Odd-Characteristic SNOVA

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Abstract. The public odd-characteristic variants of SNOVA use symmetric scalar-expanded public quadratic matrices. We show that this symmetry creates a universal quotient inside the affine-column forgery framework of Beullens. The existing analysis assigns one homogeneous feature coordinate to every ordered power pair (a, b). For symmetric S and P_i, however,

$$u^\mathsf{T}(I_n \otimes S^a)P_i(I_n \otimes S^b)u = u^\mathsf{T}(I_n \otimes S^b)P_i(I_n \otimes S^a)u.$$

Thus, with $A = \mathbb{F}_q[S]$ viewed as an ℓ-dimensional $\mathbb{F}_q$-space, the homogeneous map factors through $\mathrm{Sym}^2_{\mathbb{F}_q}(A)$ and has rank at most $m_1\binom{\ell+1}{2}$ rather than $m_1\ell^2$. In odd characteristic, the canonical quotient discards one copy of $\bigwedge^2_{\mathbb{F}_q}(A)$ per base form.

For the nine $q = 19$ parameter sets in the public Version 2.3 draft, the quotient gives residual systems ranging from 48 quadratics in 112 variables to 96 quadratics in 224 variables. Under the same semi-regular Hashimoto/Wiedemann methodology used in the SNOVA analysis, and using the stricter convention in the public analysis code, the estimated costs range from $2^{130.30}$ to $2^{238.77}$ gates. Every set is below its nominal NIST category, with estimated shortfalls from 4.06 to 44.05 bits. The same quotient also places every public Version 2.4 preview shape below target, conditional on retaining symmetric public matrices.

Fixed affine offsets bypass the verifier's rejection of symmetric signature blocks without changing the homogeneous reduction. A source-level reconstruction of the official Level-I $(28, 5, 19, 4, 4)$ KAT gives exact equivalence between the 80 verification equations and 50 quadratics in 102 variables. An independent two-column restriction gives 50 self-quadrics on a 52-dimensional affine space. These algebraic reductions and certificates are exact; the displayed work factors remain heuristic, as are the corresponding estimates in the SNOVA materials. We therefore claim a security-category or parameter break of the public odd-characteristic drafts under the candidate's accepted methodology, not a practical full-size forgery. This quotient does not apply to the unsymmetrized $q = 16$ public matrices.

Keywords: SNOVA · multivariate signatures · post-quantum cryptography · forgery attack · symmetric square · underdetermined MQ · NIST PQC

1 Introduction

SNOVA is a structured Oil-and-Vinegar signature scheme designed to combine very small public keys with short signatures and fast verification. NIST advanced SNOVA to the third round of the Additional Digital Signatures process in May 2026, while allowing the surviving candidates to submit updated specifications and implementations [12, 11]. When this manuscript was prepared, NIST's Round 3 page listed SNOVA and linked to the team website, but did not host a definitive Round 3 specification package.

The public scheme materials contain several parameter generations. The NIST-posted Round 2 package uses $\mathbb{F}_{16}$ and nonsymmetrized public base matrices. The later Version 2.3 draft introduces implemented $q = 19$ variants whose scalar-expanded public matrices are symmetric, and the public analysis repository subsequently posted six parameter shapes labeled Version 2.4 [16, 14]. The SNOVA authors have also presented odd characteristic and symmetric-form variants as a principal redesign direction [4]. The present paper analyzes that public odd-characteristic construction.

The Version 2.3 attack chapter retains a Beullens-style forgery analysis in which the quadratic feature vector is indexed by all ordered power pairs (a, b) and is assigned dimension $m_1\ell^2$. It models the induced matrix E_R as essentially random in that ordered space and derives security estimates from a rank near the ordered-coordinate maximum. The draft itself states that this chapter is inherited from Version 2.0 and still needs updating [16].

This paper shows that the ordered model is incompatible with the new public symmetry. Put $A = \mathbb{F}_q[S]$. The irreducibility of the characteristic polynomial makes A a field isomorphic to $\mathbb{F}_{q^\ell}$, while $S = S^\mathsf{T}$ makes every element of A symmetric. For fixed u and public base form P_i, the Gram pairing

$$(\xi, \eta) \longmapsto u^\mathsf{T}(I_n \otimes \xi)P_i(I_n \otimes \eta)u$$

is a symmetric $\mathbb{F}_q$-bilinear form on the ℓ-dimensional space A. It therefore factors through $\mathrm{Sym}^2_{\mathbb{F}_q}(A)$. Over odd characteristic, the ordered tensor square decomposes as

$$A \otimes_{\mathbb{F}_q} A = \mathrm{Sym}^2_{\mathbb{F}_q}(A) \oplus \bigwedge\nolimits^2_{\mathbb{F}_q}(A),$$

and the alternating summand is invisible to quadratic evaluation. In coordinates this is the equality of the (a, b) and (b, a) features. The effective feature dimension is therefore

$$K = m_1\binom{\ell + 1}{2},$$

not $m_1\ell^2$. This identity is universal for every key generated with symmetric public matrices; it does not rely on a weak-key rank drop and cannot be removed by adding more A, B, Q emulsifier terms.

1.1 Results

Our main algebraic result is the following.

Table 1. Symmetry-reduced residual systems and estimated attack costs. "Old" is the Beullens-forgery estimate in SNOVA Version 2.3 Table 11. "New" uses the stricter convention in the current analysis code. Negative margins mean that the estimate is below the category target.

Level	Parameters (v, o, ℓ, r)	Residual MQ	Old	New	Target	Margin
I	$(28, 5, 4, 4)$	50 in 102	194	138.94	143	-4.06
I	$(48, 16, 2, 2)$	48 in 112	164	130.30	143	-12.70
I	$(28, 4, 4, 5)$	50 in 98	194	138.94	143	-4.06
III	$(40, 7, 4, 4)$	70 in 146	264	184.29	207	-22.71
III	$(72, 24, 2, 2)$	72 in 168	234	184.29	207	-22.71
III	$(38, 5, 4, 5)$	70 in 142	239	184.29	207	-22.71
V	$(50, 9, 4, 4)$	90 in 182	333	227.95	272	-44.05
V	$(96, 32, 2, 2)$	96 in 224	303	238.77	272	-33.23
V	$(52, 6, 4, 6)$	90 in 178	333	227.95	272	-44.05

Theorem 1 (Universal symmetry quotient, informal). *Consider a SNOVA public key for which the power matrix S and the scalar-expanded public base matrices $P_1, \ldots, P_{m_1}$ are symmetric. After any affine-column substitution in the Beullens framework, the homogeneous quadratic part factors through*

$$\bigoplus_{i=1}^{m_1} \mathrm{Sym}^2_{\mathbb{F}_q}(\mathbb{F}_q[S]).$$

Consequently its rank is at most $K = m_1\binom{\ell+1}{2}$. If the quotient map and the induced affine constraints have their generic full ranks, then forging reduces to solving K quadratic equations in

$$N_{\mathrm{res}} = \ell(v + o) - (or\ell - K)$$

variables.

The full theorem, including nongeneric ranks, appears as theorems 3 and 4. The rank cap comes from the domain of the quadratic feature map. The A, B, Q emulsifier matrices may randomize the induced map on the symmetric quotient, but they cannot recover the alternating coordinates discarded before emulsification.

Applying the submission's own MQ methodology gives the Version 2.3 estimates in table 1. We use the more conservative of two public conventions: the current analysis code inserts one homogenizing variable, whereas the values printed in Version 2.3 Table 11 numerically match the convention without that variable. Even under the stricter convention, every $q = 19$ set falls below target. Applying the identical quotient to the Version 2.4 analysis preview gives the additional estimates in table 3.

We further show that the verifier's rejection of symmetric signature blocks does not repair the issue. In the square variants, attacker-chosen column relations and fixed offsets can force a prescribed nonzero skew entry in every generated block. Rectangular signature blocks are not symmetric objects in the first place.

The implementation audit supplies two independent exact normal forms. First, a source-level reconstruction of the official Level-I $(28, 5, 19, 4)$ KAT gives exact equivalence between the 80 verifier equations and 50 quadratics in 102 variables. Second, every one of the six individual cross-column feature maps for the same public constants is an invertible 80×80 matrix. An explicit two-column restriction fixes one column, solves 80 cross equations linearly for the second, and leaves exactly 50 self-quadrics on a 52-dimensional affine space. This second normal form is a useful public certificate of the same symmetry-assisted collapse; the headline work factor remains the conservative estimate for the rejection-safe affine-column attack.

1.2 Scope of the claim

The unconditional contribution is the algebraic reduction from ordered power pairs to a symmetric square, together with exact affine reductions and a verifier-rejection bypass. The numerical exponents use the same semi-regularity and sparse-linear-algebra assumptions employed by the SNOVA submission and prior SNOVA cryptanalysis. We have not carried out a production-size forgery at the estimated cost. The appropriate claim is therefore a *security-category or parameter break under the candidate's accepted methodology*, not a practical or unconditional running-time break.

The public versioning must also be kept explicit:
- the NIST-posted Round 2 package uses unsymmetrized $q = 16$ matrices and is not affected by this particular quotient;
- the nine $q = 19$ Version 2.3 draft sets are directly covered and have public implementations and KATs;
- the six Version 2.4 parameter shapes presently appear only in the public analysis repository; our table is conditional on any corresponding implementation retaining the symmetric public-matrix rule;
- NIST's Round 3 page listed SNOVA and linked to the team website, but did not host a definitive Round 3 specification package when this manuscript was prepared.

The submitted $q = 16$ line is outside this theorem. Recent structure-aware wedge analysis nevertheless reports below-target attacks on that line, with a rank conjecture retained in its complexity analysis [2]. Our target here is the public odd-characteristic redesign.

1.3 Organization

Section 2 fixes notation and recalls the affine-column framework. Section 3 proves the quotient. Section 4 derives the residual MQ system, and section 5 handles the verifier's rejection rule. Section 6 gives the cross-column 50-in-52 normal form. Section 7 evaluates the Version 2.3 sets and the conditional Version 2.4 preview. Section 8 reports the implementation and reproducibility checks. Section 9 discusses prior work and novelty, and section 10 gives countermeasures. The appendices collect parameter formulas, the coordinate form of the quotient, and a claim checklist.

2 Preliminaries and the Existing Forgery Framework

2.1 Parameters and scalar expansion

Let q be a prime power, let ℓ be the size of the square matrix blocks, and let r be the number of columns of each signature block. The public map has $n = v + o$ matrix variables. Version 2.3 defines

$$m_1 = \left\lceil \frac{or}{\ell} \right\rceil \quad \text{and} \quad M = or\ell,$$

where m_1 is the number of scalar-expanded public base matrices and M is the number of scalar output equations. A signature is an n-tuple of blocks in $\mathrm{Mat}_{\ell\times r}(\mathbb{F}_q)$. Equivalently, writing its columns as

$$U = [u_0 \mid u_1 \mid \cdots \mid u_{r-1}], \qquad u_j \in \mathbb{F}_q^{n\ell},$$

the attack starts with $n\ell$ variables after expressing the other columns affinely in terms of u_0.

Let $S \in \mathrm{Mat}_\ell(\mathbb{F}_q)$ be the public symmetric matrix with irreducible characteristic polynomial, and put

$$\Sigma_a = I_n \otimes S^a \in \mathrm{Mat}_{n\ell}(\mathbb{F}_q), \qquad 0 \le a < \ell.$$

For each base matrix $P_i \in \mathrm{Mat}_{n\ell}(\mathbb{F}_q)$ define the bilinear power-pair forms

$$B_i^{(a,b)}(x, y) = x^\mathsf{T} \Sigma_a P_i \Sigma_b y. \tag{1}$$

The public odd-characteristic algorithms generate each scalar-expanded public matrix symmetrically: diagonal blocks are symmetric and opposite off-diagonal blocks are transposes. Thus $P_i^\mathsf{T} = P_i$ for the public $q = 19$ drafts.

2.2 The Beullens affine-column substitution

Beullens observed that the SNOVA public map can be written as a linear transformation of the values $B_i^{(a,b)}(u_j, u_k)$ [1]. The attacker selects matrices $R_j \in \mathbb{F}_q[S]$ and offsets $v_j \in \mathbb{F}_q^{n\ell}$, with $R_0 = I$, and substitutes

$$u_j = (I_n \otimes R_j)u + v_j, \qquad 0 \le j < r. \tag{2}$$

after identifying $u = u_0$ and taking $v_0 = 0$ if desired.

For a fixed relation $R = (R_0, \dots, R_{r-1})$, the substituted verification map for a target $t \in \mathbb{F}_q^M$ can be written

$$E_R z_{\mathrm{ord}}(u) + L_R u + c_R = t. \tag{3}$$

Here
$$z_{\mathrm{ord}}(u) = \Big(B_i^{(a,b)}(u,u)\Big)_{1\le i\le m_1,\ 0\le a,b<\ell} \in \mathbb{F}_q^{m_1\ell^2},$$
$E_R \in \mathbb{F}_q^{M\times m_1\ell^2}$ depends only on public constants and the chosen column relation, while $L_R u + c_R$ collects all cross and constant terms involving the offsets.

In the ordered-coordinate analysis, if $\operatorname{rank}(E_R) = \rho$, Gaussian elimination supplies $M-\rho$ affine-linear equations and leaves at most ρ independent quadratic equations. The Version 2.3 attack chapter models E_R as essentially random and bases its concrete estimate on ρ near the ordered-coordinate maximum. Our observation is that, whenever the public base matrices are symmetric, E_R is applied to a feature vector with deterministic coordinate equalities.

3 The Symmetry Quotient

Lemma 1 (Power-pair symmetry). *Suppose $S^\mathsf{T} = S$ and $P_i^\mathsf{T} = P_i$. Then for every $u \in \mathbb{F}_q^{n\ell}$ and every $0 \le a, b < \ell$,*
$$B_i^{(a,b)}(u,u) = B_i^{(b,a)}(u,u).$$

Proof. The quantity is a scalar, so it equals its transpose:
$$\begin{aligned} u^\mathsf{T}\Sigma_a P_i \Sigma_b u &= \left(u^\mathsf{T}\Sigma_a P_i \Sigma_b u\right)^\mathsf{T} \\ &= u^\mathsf{T}\Sigma_b^\mathsf{T} P_i^\mathsf{T} \Sigma_a^\mathsf{T} u = u^\mathsf{T}\Sigma_b P_i \Sigma_a u. \end{aligned}$$
We used that every power of the symmetric matrix S is symmetric.

Theorem 2 (Structural symmetric-square factorization). *Let*
$$A = \mathbb{F}_q[S] \cong \mathbb{F}_{q^\ell}$$
be viewed as an ℓ-dimensional vector space over $\mathbb{F}_q$. For a fixed $u \in \mathbb{F}_q^{n\ell}$ and public base matrix P_i, define
$$\Phi_{u,i}(\xi,\eta) = u^\mathsf{T}(I_n \otimes \xi)P_i(I_n \otimes \eta)u, \qquad \xi,\eta \in A.$$
Then $\Phi_{u,i}$ is a symmetric $\mathbb{F}_q$-bilinear form on A. Consequently it factors uniquely through $\mathrm{Sym}^2_{\mathbb{F}_q}(A)$, and the combined homogeneous feature map for $P_1,\dots,P_{m_1}$ factors through
$$\bigoplus_{i=1}^{m_1} \mathrm{Sym}^2_{\mathbb{F}_q}(A).$$
When the characteristic is odd, the kernel of the canonical ordered-to-symmetric quotient is canonically
$$\bigoplus_{i=1}^{m_1} \bigwedge\nolimits^2_{\mathbb{F}_q}(A),$$
of dimension $m_1\binom{\ell}{2}$.

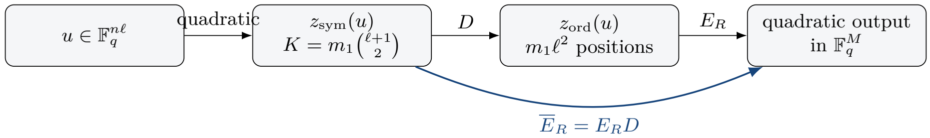

Fig. 1. The ordered feature vector lies in the image of a K-dimensional duplication map. Randomizing E_R cannot increase the rank of the composite beyond K.

Proof. The irreducibility of the characteristic polynomial makes A a field, and $S = S^{\mathsf{T}}$ makes every element of A symmetric. Bilinearity over $\mathbb{F}_q$ is immediate. For $\xi, \eta \in A$, the value is a scalar and hence is equal to its transpose:

$$\begin{aligned}
\Phi_{u,i}(\eta, \xi) &= u^{\mathsf{T}}(I_n \otimes \eta)P_i(I_n \otimes \xi)u \\
&= \big(u^{\mathsf{T}}(I_n \otimes \eta)P_i(I_n \otimes \xi)u\big)^{\mathsf{T}} \\
&= u^{\mathsf{T}}(I_n \otimes \xi)P_i(I_n \otimes \eta)u = \Phi_{u,i}(\xi, \eta).
\end{aligned}$$

Thus $\Phi_{u,i}$ is symmetric and factors through the universal symmetric square. In odd characteristic, $A \otimes_{\mathbb{F}_q} A = \mathrm{Sym}^2_{\mathbb{F}_q}(A) \oplus \bigwedge^2_{\mathbb{F}_q}(A)$, so the alternating summand is exactly the discarded part.

This proof explains the numerical loss. For $\ell = 4$, the ordered power space has dimension 16, while its symmetric square has dimension 10; the six missing directions are exactly $\bigwedge^2(A)$. With $m_1 = 5$, this gives the 30 affine-linear output directions observed in the Level-I end-to-end experiment.

Define the unordered feature vector

$$z_{\text{sym}}(u) = \Big(B_i^{(a,b)}(u,u)\Big)_{1 \le i \le m_1,\ 0 \le a \le b < \ell} \in \mathbb{F}_q^K, \qquad K = m_1\binom{\ell+1}{2}.$$

Let

$$D : \mathbb{F}_q^K \longrightarrow \mathbb{F}_q^{m_1\ell^2}$$

be the duplication map that sends an unordered coordinate $(i, \{a, b\})$ to both ordered positions (i, a, b) and (i, b, a) when $a \neq b$, and to the single diagonal position when $a = b$. Then

$$z_{\text{ord}}(u) = D z_{\text{sym}}(u). \tag{4}$$

Theorem 3 (Symmetry rank bound). *For every public key whose power matrix and scalar-expanded public base matrices are symmetric, and for every affine-column relation R, the following holds. The homogeneous quadratic part in equation (3) factors as*

$$\overline{E}_R z_{\text{sym}}(u), \qquad \overline{E}_R = E_R D \in \mathbb{F}_q^{M \times K}.$$

In particular,

$$\mathrm{rank}(\overline{E}_R) \le K = m_1\binom{\ell+1}{2}.$$

Proof. The structural factorization is theorem 2. In the power basis $1, S, \ldots, S^{\ell-1}$, it becomes the coordinate identity equation (4). Substituting that identity into equation (3) gives the displayed factorization, and the rank bound follows from the number of columns of $\overline{E}_R$.

Why the emulsifier does not remove the relation. The Version 2.3 changes use many independently generated $A_{i,\alpha}, B_{i,\alpha}, Q_{i,\alpha,1}, Q_{i,\alpha,2}$ matrices so that the ordered-coordinate E_R resembles a general matrix rather than the repeated block-diagonal matrix attacked in the first-round construction. This may make $\overline{E}_R$ a generic map on its K-dimensional domain. It cannot make that domain larger. The lost antisymmetric coordinates vanish before the emulsifier is applied.

The reduction is universal, not a weak-key event. Prior analyses search over relations R to find an unusually low rank. Here the cap K holds for every relation and every public key satisfying the public-matrix symmetry rule. A further rank drop below K can only improve the residual system relative to the full-rank quotient analyzed below.

4 The Symmetry-Assisted Forgery Attack

Let

$$\rho = \operatorname{rank}(\overline{E}_R) \leq K.$$

Choose a full-rank matrix $W_R \in \mathbb{F}_q^{(M-\rho)\times M}$ whose rows span the left kernel of $\overline{E}_R$. Multiplying equation (3) by W_R eliminates every quadratic term:

$$W_R L_R u = W_R(t - c_R). \tag{5}$$

Let $\lambda_R = \operatorname{rank}(W_R L_R)$. Gaussian elimination expresses λ_R coordinates of u as affine functions of the remaining

$$N_R = n\ell - \lambda_R$$

coordinates. Substitution into a row basis of equation (3) leaves at most ρ quadratic equations.

Theorem 4 (Residual MQ reduction). *For any relation and offsets, forging a selected target reduces in polynomial time to solving at most ρ quadratic equations in $n\ell - \lambda_R$ variables, where*

$$\rho \leq m_1 \binom{\ell+1}{2}, \qquad \lambda_R \leq M - \rho.$$

If both ranks are generic, namely $\rho = K$ and $\lambda_R = M - K$, the residual system consists of

$$\boxed{K = m_1 \binom{\ell+1}{2} \text{ quadratics in } N_{\mathrm{res}} = n\ell - M + K \text{ variables.}} \tag{6}$$

A solution extends by linear algebra to a signature satisfying the original M verification equations.

Proof. The left-kernel step gives equation (5). Solve a maximal independent subsystem and substitute into the original equations. Modulo the affine constraints, the quadratic coefficient vectors lie in the ρ-dimensional column space of $\overline{E}_R$, so a row basis gives at most ρ independent quadratics. Reversing the substitution produces u, and equation (2) gives all signature columns. Because the reduction used row operations and exact substitutions, the resulting signature satisfies all M original equations.

Underdetermination is preserved. When $\lambda_R = M - \rho$, the difference between variables and quadratic equations is

$$(n\ell - M + \rho) - \rho = n\ell - M,$$

independent of ρ. Thus a further quotient-rank defect decreases the equation count without changing the nominal underdetermination. The full-rank quotient $\rho = K$ is the natural universal case and is the one observed in our experiments.

Attack algorithm. For an arbitrary message $\mathcal{M}$:
1. sample a salt and compute the target $t = H(\mathcal{M}\|\mathsf{salt})$;
2. choose a public column relation R and offsets v_j;
3. construct $\overline{E}_R, L_R, c_R$ and row-reduce as above;
4. solve the residual underdetermined MQ system;
5. reconstruct u and then the complete signature matrix U using equation (2).

The first, second, fourth, and fifth steps are the same attack architecture used by Beullens. The new ingredient is that step 3 must quotient the homogeneous features by symmetry before estimating their rank.

5 Bypassing the Symmetric-Block Rejection

Version 2.3 notes that signatures composed of powers of S can be dangerous when the public matrices are symmetric. For odd q, the verifier therefore rejects square signature blocks that are symmetric: it allows none when $\ell > 2$ and at most $\lfloor n/4 \rfloor$ when $\ell = 2$.

The affine-column attack has fixed offsets in addition to its free vector u. Those offsets can enforce nonsymmetry identically.

Lemma 2 (A constant-skew column relation). *Assume that the characteristic polynomial of S is irreducible. Let e_0, e_1 be two distinct standard basis vectors of $\mathbb{F}_q^\ell$. There exists $R_1 \in \mathbb{F}_q[S]$ such that*

$$e_0^\mathsf{T} R_1 = e_1^\mathsf{T}.$$

With $R_0 = I$, choose offsets for the first two signature columns so that, in each matrix block,

$$(v_1)_0 - (v_0)_1 = \delta$$

for a fixed $\delta \in \mathbb{F}_q^\times$. Then every assignment to u satisfies

$$U[0,1] - U[1,0] = \delta$$

in that block. In particular, the block is never symmetric.

Proof. Because the characteristic polynomial is irreducible, $\mathbb{F}_q^\ell$ is a simple $\mathbb{F}_q[S]$-module. Hence the nonzero row vector $e_0^\top$ is cyclic under right multiplication by $\mathbb{F}_q[S]$, and some R_1 maps it to $e_1^\top$. Under the relation $u_0 = u + v_0$ and $u_1 = (I_n \otimes R_1)u + v_1$, the variable-dependent parts of the two entries are $e_1^\top u$ and $e_0^\top R_1 u$, which coincide. Their difference is therefore the chosen offset constant δ.

The construction can be applied independently to every block. It does not change the homogeneous quadratic map $\overline{E}_R$; offsets affect only L_R and c_R. For the square $\ell = 2$ sets it makes every block nonsymmetric, which is stronger than the verifier's threshold. For the square $\ell = 4$ sets it satisfies the zero-symmetric-block requirement. The rectangular variants have blocks in $\mathrm{Mat}_{\ell\times r}$ with $r \neq \ell$, so matrix symmetry is not defined and no such rejection is applied.

6 A Cross-Column Gram Normal Form

The official Level-I square parameters admit a second exact public reduction. Write the four signature columns as $x_0, x_1, x_2, x_3 \in \mathbb{F}_{19}^{132}$. For one column define the self-feature vector

$$Z_s(x_s) = \big(B_i^{(a,b)}(x_s, x_s)\big)_{1\le i\le 5,\ 0\le a\le b<4} \in \mathbb{F}_{19}^{50},$$

and, for $s < t$, define the ordered cross-feature vector

$$G_{st}(x_s, x_t) = \big(B_i^{(a,b)}(x_s, x_t)\big)_{1\le i\le 5,\ 0\le a,b<4} \in \mathbb{F}_{19}^{80}.$$

Let $E_{ss} : \mathbb{F}_{19}^{50} \to \mathbb{F}_{19}^{80}$ and $E_{st} : \mathbb{F}_{19}^{80} \to \mathbb{F}_{19}^{80}$ denote the corresponding public output maps determined by the official A, B, Q constants.

Proposition 1 (Official Level-I feature-map certificate). *For the Version 2.3 Level-I parameters $(v, o, q, \ell, r) = (28, 5, 19, 4, 4)$:*
1. *each self-column map E_{ss} has rank 50;*
2. *each of the six cross-column maps E_{st} is invertible;*
3. *the full 80×680 symmetry-reduced feature map has rank 80.*

These are exact finite-field rank statements for the official public constants.

Restrict the verifier to $(x_0, x_1, 0, 0)$. Its output can be written

$$F(x_0, x_1) = E_{00}Z_0(x_0) + E_{01}G_{01}(x_0, x_1) + E_{11}Z_1(x_1). \tag{7}$$

The artifact contains an explicit public vector x_0 for which the linear map

$$C_{x_0} : x_1 \longmapsto E_{01}G_{01}(x_0, x_1)$$

has rank 80. For a target $t \in \mathbb{F}_{19}^{80}$, first solve the affine linear system

$$C_{x_0}(x_1) = t - E_{00}Z_0(x_0). \tag{8}$$

Its solution set is an affine space of dimension $132 - 80 = 52$. On that affine space, equation (7) equals t exactly when

$$E_{11}Z_1(x_1) = 0.$$

Because E_{11} has column rank 50, this is precisely

$$\boxed{\text{50 quadratic equations in 52 affine variables over } \mathbb{F}_{19}.}$$

Thus any solution reconstructs a preimage of the original 80-coordinate verifier. The public artifact verifies the feature-map ranks, the rank-80 coefficient matrix for the displayed x_0, and direct equality with the official verifier.

This two-column normal form is logically independent of the one-base left-kernel reduction in section 4. It is an exact algebraic verifier-preimage normal form; accepted signatures must additionally satisfy the verifier's format check. Accordingly, we do not use it to lower the headline complexity estimate: the conservative Hashimoto analysis of the rejection-safe 50-in-102 family already specializes most surplus variables before its square solve.

7 Concrete Complexity Estimates

7.1 Methodology

We deliberately evaluate the new residual systems with the methodology already used in the SNOVA submission and in Beullens's paper, rather than introducing a more aggressive cost model. For a determined or overdetermined MQ system, the model estimates sparse Wiedemann/FXL from a semi-regular operating degree. Hashimoto's method reduces an underdetermined system of m equations in n variables to a collection of smaller determined systems, optimizing over its parameters (a, k) [5, 1].

Version 2.3 gives the Beullens cost as

$$\mathsf{Hashimoto}(n\ell - \ell^2 m_1 + R, R, q)$$

up to its field-to-gate conversion. We replace $(R, n\ell - \ell^2 m_1 + R)$ by the symmetry-reduced values $(K, n\ell - M + K)$ from equation (6).

There is a small convention discrepancy in the public materials. The values in Version 2.3 Table 11 are numerically reproduced by using no extra homogenizing variable in the semi-regular series. The current analysis script passes one extra variable. The latter increases the new estimates by about 4.25 bits in most rows. We use that stricter convention in the headline table and give both conventions in table 2.

Table 2. Estimates under both semi-regular conventions. The $p_1 = 1$ column is the stricter current analysis-code convention and is used in the abstract. The $p_1 = 0$ column numerically matches the convention reflected in Version 2.3 Table 11.

Level	Parameters	Residual MQ	$p_1 = 0$	$p_1 = 1$	Target
I	$(28, 5, 4, 4)$	$50/102$	134.69	138.94	143
I	$(48, 16, 2, 2)$	$48/112$	126.05	130.30	143
I	$(28, 4, 4, 5)$	$50/98$	134.69	138.94	143
III	$(40, 7, 4, 4)$	$70/146$	180.05	184.29	207
III	$(72, 24, 2, 2)$	$72/168$	181.89	184.29	207
III	$(38, 5, 4, 5)$	$70/142$	180.05	184.29	207
V	$(50, 9, 4, 4)$	$90/182$	223.70	227.95	272
V	$(96, 32, 2, 2)$	$96/224$	234.52	238.77	272
V	$(52, 6, 4, 6)$	$90/178$	223.70	227.95	272

Table 3. Symmetry-reduced estimates for the Version 2.4 analysis preview. These are not definitive Round 3 parameters; the table evaluates the public analysis-repository shapes under the same $p_1 = 1$ convention used in the headline Version 2.3 table.

Level	Parameters (v, o, ℓ, r)	Residual MQ	New	Target	Margin
I	$(26, 5, 4, 4)$	$50/94$	138.94	143	-4.06
I	$(43, 17, 2, 2)$	$51/103$	141.57	143	-1.43
III	$(37, 7, 4, 4)$	$70/134$	184.29	207	-22.71
III	$(64, 25, 2, 2)$	$75/153$	194.57	207	-12.43
V	$(44, 6, 4, 6)$	$90/146$	227.95	272	-44.05
V	$(90, 33, 2, 2)$	$99/213$	246.84	272	-25.16

7.2 All odd-characteristic parameter sets

For the nine $q = 19$ parameter shapes in Version 2.3, equation (6) gives the residual systems shown in table 2. The category targets are the values used by the submission: 143, 207, and 272 bits for Levels I, III, and V.

The Level-I $\ell = 4$ rows have the narrowest margin: 4.06 bits under the stricter convention. We therefore describe them as *estimated below target*, not as an unconditional violation. The remaining rows have margins of 12.70 to 44.05 bits under the same convention.

7.3 Version 2.4 analysis preview

The public SNOVA analysis repository was updated with six parameter shapes labeled Version 2.4, although no matching specification or implementation package was public at the time of this audit [14]. The symmetry quotient is algebraic and does not depend on the numerical parameters. If the preview retains the symmetric public-matrix construction, the stricter $p_1 = 1$ convention gives table 3.

The first Level-I margin is the same narrow 4.06-bit estimate as in Version 2.3, and the second is only 1.43 bits. Those two rows should be described as estimated shortfalls. The Level-III and Level-V margins are substantially larger. Reparameterizing within the same symmetric construction must therefore use $m_1\binom{\ell+1}{2}$, not $m_1\ell^2$, as the homogeneous rank ceiling.

7.4 Interpretation of the estimates

The estimates inherit three standard assumptions:

1. the residual systems behave close enough to semi-regular systems for the operating-degree calculation to be meaningful;
2. sparse linear algebra has the costs assumed by the submission's model;
3. field operations are converted to gates using the same coarse formula as the existing analysis.

These assumptions are neither introduced nor proved by this work. They are the methodology under which Version 2.3 claims its security levels. Our conclusion is therefore comparative: once the homogeneous feature dimension is corrected, the candidate's own framework places the odd-characteristic sets below their categories.

The Cabarcas–Li–Verbel–Villanueva-Polanco solver exploits additional multi-graded structure in SNOVA systems [3]. We do not include any further speedup from that method. Such structure could lower the residual-system costs, but is not needed for the claimed parameter shortfalls.

8 Implementation and Algebraic Validation

We performed five complementary checks.

Universal identity and quotient map. A self-contained checker verifies lemma 1 over sampled symmetric matrices and constructs the duplication map explicitly. For $(m_1, \ell) = (5, 4)$ the ordered feature dimension is 80, the unordered feature dimension is 50, and the duplication map has rank 50.

All-parameter quotient ranks. Using the fixed public A, B, Q constants specified by Version 2.3 and the public relation $R_0 = I$, $R_j = 0$ for $j > 0$, the quotient matrices attain full column rank for all nine $q = 19$ shapes. Their matrix sizes and ranks are exactly the values implicit in table 4. On the Level-I $\ell = 4$ constants, all $19^3 = 6{,}859$ scalar projective relations tested produced an 80×50 quotient matrix of rank 50, and 10,000 sampled A-valued relations did the same. Full quotient rank is the conservative case used in the cost table.

Official-key reconstruction. An independent Python transcription of the relevant Version 2.3 key-generation routines reconstructs the Level-I $(28, 5, 19, 4)$ KAT public key byte-for-byte from the KAT secret-key record. This checks the field arithmetic, fixed A, B, Q generation, documented coefficient repair, public-matrix expansion, and public-key packing used in the attack harness.

End-to-end verifier equivalence and format bypass. On the KAT-derived key and two independently generated keys, the affine-column system gives

$$\operatorname{rank}(\overline{E}_R) = 50, \qquad \dim\ker(\overline{E}_R^{\mathsf{T}}) = 30, \qquad \operatorname{rank}(W_R L_R) = 30.$$

Eliminating the 30 affine equations leaves 50 quadratics in 102 variables. A known full-system solution is preserved, and the selected 50 residual output coordinates reconstruct all 80 original verification coordinates exactly. The same harness verifies algebraically that the fixed-offset construction of lemma 2 makes every square signature block nonsymmetric for every residual assignment.

Cross-column certificate. The complete Level-I symmetry-reduced feature map has size 80×680 and rank 80. Each self-column submap has rank 50, each cross-column submap is an invertible 80×80 matrix, and the explicit x_0 used in equation (8) gives an 80×132 coefficient matrix of rank 80. Restricting the 50 self quadrics to its affine kernel preserves a 50-dimensional quadratic span, confirming the exact 50-in-52 formulation.

8.1 Public reproducibility artifact

The accompanying artifact contains:
- the source-level Version 2.3/KAT reconstruction and end-to-end verifier-equivalence checker;
- an independent implementation of both semi-regular conventions and the Hashimoto optimization;
- a standalone cross-column certificate checker;
- machine-readable rank, elimination, cost, and cross-column results;
- the Version 2.3 specification used in the audit and a SHA-256 manifest.

The artifact pins the audited Version 2.3 repository state and explains how to generate the official KAT from that state before running the byte-level reconstruction. Python 3 and NumPy suffice for the core checks.

9 Related Work and Novelty

Beullens's forgery attack. Beullens identified SNOVA's structured whipping layer, expressed the public map through ordered bilinear power-pair coordinates, and showed that low-rank combinations of the corresponding emulsifier matrices lead to forgery attacks [1]. The Version 2.3 countermeasure increases the number and diversity of A, B, Q terms to make the ordered-coordinate E_R resemble a random matrix.

Our attack uses the same affine-column architecture but a different source of rank loss. We do not find a low-rank element in a family of otherwise full-rank ordered-coordinate matrices. We show that the odd-characteristic homogeneous feature vector itself lies in a K-dimensional symmetric subspace. Thus the quotient applies to every key and every relation, even when the induced map is maximally random and full rank on that quotient.

Stable-ideal attacks. Cabarcas et al. exploit the action of the commutative group generated by S to design a polynomial-system solver specialized to SNOVA [3]. Their work can be viewed as improving the residual-system solving stage. The present result instead reduces the number of residual equations before solving begins. The approaches are compatible, but we do not combine their cost benefits here.

Characteristic-two wedge attacks and the odd-characteristic redesign. The structure-aware wedge attack exploits SNOVA's block-ring structure and reports below-target attacks on the NIST Round 2 $q = 16$ parameters, while explicitly retaining a conjectural rank component in the complexity analysis [2]. The SNOVA authors subsequently investigated the wedge map and proposed a reformulation emphasizing odd-characteristic and symmetric-form variants [15, 4]. Our attack is complementary: it does not apply to the unsymmetrized $q = 16$ public matrices, but it attacks the public odd-characteristic repair direction through an elementary quotient that is exact and independent of the wedge-rank conjecture.

Key-recovery attacks. Several works interpret the scalar expansion of SNOVA as UOV, or lift it to smaller extension-field UOV instances, and transfer key-recovery attacks [6, 9, 10]. Our result is a universal forgery reduction and does not recover the secret oil space.

Symmetric-algebra lineage. The decomposition $V \otimes V = \mathrm{Sym}^2(V) \oplus \bigwedge^2(V)$ and the fact that a quadratic evaluation kills the alternating part are classical; we claim no novelty for that algebra [8]. The closest cryptanalytic precedent is the p-reduced symmetric-algebra framework of Jin, Pan, He, Gong, and Ding [7], which transports the characteristic-two wedge attack of Ran [13] to odd characteristic and emphasizes that the effective monomial algebra can be smaller than a characteristic-blind model. Our use is different and more elementary: the public symmetry introduced specifically in SNOVA 2.3 forces the Beullens feature map itself to factor through a symmetric square. Association-scheme or Delsarte machinery concerns related objects but is not the mechanism of the attack and is not needed here.

Novelty boundary. The contribution is recognizing that the public odd-characteristic redesign makes this elementary quotient load-bearing in the existing SNOVA affine-column attack; deriving the exact rank and residual-system consequences for square and rectangular variants; giving the structural $\mathrm{Sym}^2(A)$ proof and the exact two-column normal form; proving that the advertised signature rejection does not block the attack; and validating the reduction against the current implementation and cost methodology. We found no corresponding symmetry quotient in the Version 2.3 attack chapter or the Version 2.4 analysis preview. The former continues to use ordered ℓ^2 coordinates and explicitly states that it still requires updating.

10 Countermeasures and Standardization Implications

The immediate requirement is to evaluate every symmetric-matrix parameter set using

$$K = m_1 \binom{\ell + 1}{2}$$

as the maximum homogeneous rank in the affine-column attack. Merely choosing A, B, Q matrices for which the ordered-coordinate E_R has high rank cannot overcome the quotient.

Possible design responses include:

1. **Reparameterization.** Increase the dimensions until the symmetry-reduced residual MQ system meets the desired category under the team's accepted cost methodology.
2. **Remove public-matrix symmetry.** This sacrifices the key-size reduction that motivated the odd-characteristic change, but removes lemma 1 in its present form.
3. **Redesign the whipping layer.** A construction with a provably full-rank emulsifier on its actual quadratic feature space may prevent this style of dimensional collapse. Such a redesign requires a fresh analysis; simply randomizing more A, B, Q terms is insufficient.

The symmetric-block rejection should not be counted as a countermeasure against the affine-column attack, because fixed offsets can satisfy it identically. A rejection rule capable of blocking the attack would need to constrain the affine family itself without destroying correct signing, rather than test a property that the attacker can set with constants.

NIST advanced SNOVA to Round 3 while allowing updated specifications and implementations [11]. The Version 2.3 draft and Version 2.4 analysis preview are therefore at an appropriate stage for correcting the attack model. Returning unchanged to the submitted $q = 16$ construction is not an obvious remedy, because structure-aware wedge analysis reports below-target attacks on those parameters, subject to its stated rank conjecture. Any viable repair must address both attack families and be analyzed on the actual, symmetry-reduced feature space.

11 Conclusion

The public odd-characteristic SNOVA construction makes its scalar-expanded base matrices symmetric. That key-size optimization identifies the ordered power-pair coordinates used by the Beullens affine-column attack. The homogeneous feature map factors through

$$\bigoplus_{i=1}^{m_1} \mathrm{Sym}^2_{\mathbb{F}_q}(\mathbb{F}_q[S])$$

and has rank at most $m_1\binom{\ell+1}{2}$, not $m_1 \ell^2$. This is a universal algebraic identity, not a weak-key phenomenon.

Under the candidate's own semi-regular Hashimoto/Wiedemann framework, every $q = 19$ Version 2.3 set is estimated below its intended category. The same is true for each Version 2.4 preview shape if a corresponding implementation retains symmetric public matrices. The Level-III and Level-V shortfalls are substantial; the narrow Level-I rows should be described as estimated shortfalls. The verifier's symmetric-block rejection is bypassed by fixed skew offsets. The official Level-I implementation admits two exact public normal forms: 50 quadratics in 102 variables through affine-column elimination, and 50 self-quadrics on a 52-dimensional affine space through the cross-column certificate.

The submitted $q = 16$ package is outside this particular quotient because its public matrices are not symmetrized. The present result targets the public odd-characteristic redesign. The remaining uncertainty is the cost of solving the residual MQ systems, not the quotient or the public reductions: the algebraic attack is exact, whereas the displayed gate exponents inherit the heuristic methodology used by the SNOVA submission.

A Parameter Formulas

For convenience, table 4 records the intermediate dimensions. Recall

$$n = v + o, \quad m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \quad M = or\ell, \quad K = m_1 \binom{\ell+1}{2}, \quad N_{\mathrm{res}} = n\ell - M + K.$$

Table 4. Dimension calculation for all $q = 19$ Version 2.3 draft sets.

Level	(v, o, ℓ, r)	m_1	M	K	N_{res}
I	$(28, 5, 4, 4)$	5	80	50	102
I	$(48, 16, 2, 2)$	16	64	48	112
I	$(28, 4, 4, 5)$	5	80	50	98
III	$(40, 7, 4, 4)$	7	112	70	146
III	$(72, 24, 2, 2)$	24	96	72	168
III	$(38, 5, 4, 5)$	7	100	70	142
V	$(50, 9, 4, 4)$	9	144	90	182
V	$(96, 32, 2, 2)$	32	128	96	224
V	$(52, 6, 4, 6)$	9	144	90	178

B A Linear-Algebra View of the Quotient

For one base form, order the coordinates of $\mathbb{F}_q^{\ell^2}$ by pairs (a, b). Order the coordinates of $\mathrm{Sym}^2(\mathbb{F}_q^{\ell})$ by pairs $a \leq b$. The duplication matrix D_ℓ has one 1 in row

(a, a) and, for $a < b$, one 1 in each of rows (a, b) and (b, a). It has full column rank $\binom{\ell+1}{2}$ over every field. For m_1 base forms,

$$D = I_{m_1} \otimes D_\ell.$$

The attack matrix used in the odd-characteristic homogeneous system is $E_R D$. In particular, even an ordered-coordinate matrix E_R of full row and column rank cannot produce rank above $\operatorname{rank}(D) = K$.

For $\ell = 4$ the single-form duplication matrix maps ten coordinates to sixteen positions. The six missing dimensions are precisely the antisymmetric differences $e_{a,b} - e_{b,a}$ for $a < b$. Across $m_1 = o$ square instances, this removes $6o$ homogeneous dimensions. For the Level-I $o = 5$ set, it removes 30 dimensions, exactly the number of affine-linear equations observed in the end-to-end experiment.

C Reproducibility and Claim Checklist

A public description of this work should distinguish the following statements.

Proved algebraically. The homogeneous affine-column system factors through unordered power pairs, so its rank is at most K. Left-kernel elimination gives theorem 4, and fixed skew offsets bypass the symmetric-block rejection.

Certified computationally. The Level-I official-key reconstruction matches the KAT public key byte-for-byte; the end-to-end affine reduction is exactly equivalent to the original verifier; all six official Level-I cross-column maps are invertible; and the supplied certificate gives the 50-in-52 normal form.

Estimated heuristically. The gate complexities in tables 1 to 3, like the submission's estimates, assume semi-regular behavior and a particular sparse-linear-algebra model.

Not claimed. We do not claim a practical production-size forgery, key recovery, a break of the unsymmetrized $q = 16$ instances by this quotient, a final Round 3 parameter set, or a proven running-time upper bound matching the displayed estimates.

The SHA-256 digest of the Version 2.3 PDF used in the implementation audit is

`963fa43d7e6ed8ecf1e77556e7cf47f62d4c34177efece2cd9a9517d0fbaf783`.

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